

**DIPLOMA EXAMINATION IN
ENGINEERING/TECHNOLOGY/ MANAGEMENT/
COMMERCIAL PRACTICE.
5012- Design of Steel & RCC Structures**

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence. (9 x 1 = 9 Marks)

Module Outcome Cognitive level

1	Compression element in truss is called.....	M 1.01	R
2	The ratio of throat thickness to the size of 45° weld.....	M 1.02	R
3	Long span heavy steel beams which under high BM & SF is called	M 2.01	R
4	The cross sections in which the element buckle locally even before reaching yield stress.....	M 2.02	R
5	The maximum permissible strain in concrete in bending	M 3.04	R
6	Reinforcement provided to resist temperature cracks & cracks due to expansion & contraction.....	M 3.02	R
7	Minimum grade required for RCC as per IS 456.....	M 3.01	R
8	Minimum diameter required for longitudinal bars in column.....	M 4.01	R
9	Minimum clear cover required for a footing	M4.04	R

PART B

II. Answer any Eight questions from the following

(8 x 3 = 24 Marks)

Module Outcome Cognitive level

1	List the type of weld with figure	M 1.01	R
2	List & explain the type of failure in Tension Member	M 1.02	U
3	What are the steel sections used as a compression member	M 1.03	R
4	Differentiate laterally supported & unsupported beam	M 2.03	R
5	What are the different type of cross sections as per IS 800	M 2.02	R
6	Write the assumptions in the flexural design as per IS 456	M 3.01	R
7	Differentiate Under reinforced, over reinforced & balanced sections. Compare the depth of neutral axis of these.	M 3.02	U
8	Define development length. How the bond stress of M20 concrete modified for HYSD bars in compression as per IS 456.	M 3.03	A
9	Discuss the types of footings	M 4.03	R
10	What are the maximum spacing & min diameter for ties in column as IS standards	M 4.02	R

PART C

III. Answer all questions from the following

(6 x 7 = 42 Marks)

Module Outcome Cognitive level

1. A tie member consisting of ISA 80 x 50 x 8mm (Fe410 Grade) is welded to a 12mm thick gusset plate at site. Design weld to transmit load equal to the design strength of the member OR	M1.02	U
2. Explain about lacing system with figure. What are the codal provision for lacing system?	M1.04	U
3. Design a simply supported laterally restrained beam of span 6m under a BM of 150kNm & SF of 210kN. Assume Fe410 grade steel. Check for deflection in needed. OR	M2.03	U
4. Identify & explain the parts of plate girder with figure	M2.04	R
5. Design an RC beam 250 x 400 mm under DL 5kN/m , LL = 10kN/m & a concentrated load of 30kN at the center, beam having effective length 6m. Use M25 concrete & Fe 415 steel. Assume beam subjected to moderate exposure condition. $F_{sc} = .84f_y$ OR	M3.03	A
6. Design the shear reinforcement for the beam having dimension 250x400mm. Clearcover 30mm, beam is reinforced on the tension side with 3 No's of 20mm bars. Fe 415 steel & M 20 concrete. The working SF is 100kN.	M3.04	A
7. Design a square column to carry an axial load of 1200 kN. Use Fe 415 steel & M20 concrete. Assume as a short column OR	M4.02	U
8. Design an isolated footing for a square column 450 x 450 mm, reinforced with 8 -25# bars, and carrying a load of 2300kN. Assume soil with SBC (gross) 300kN/m ² at a depth of 1.5m below the ground. Assuming M20 & Fe 415 for the footing &	M4.04	A

M25 & Fe 415 for the column. P = 2300kN $q_a = 300\text{kN/m}^2$		
9. Design a single angle strut connected to the gusset plate to carry 180 kN factored load. The length of the strut between center to center connection is 3 m OR 10. i) Differentiate one way slab & two way slab ii) Identify the different type of two way slabs with figure, depends upon end conditions as per IS 456	M1.02	U
	M3.04	U
11. Determine the design bending strength of a laterally supported beam of ISMB 300 @ 442 N/m, the yield stress of steel is 250 MPa OR 12. i) Differentiate short column & long column ii) Write the effective length of column as per IS 456	M2.01	U
	M4.01	U